



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
UNIT TEST
03-04-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TOAS, TNAS & TAAS]

Time : 3 Hours

PAPER-1

Maximum Marks : 186

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **36** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- **Boltzmann constant** $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant** $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

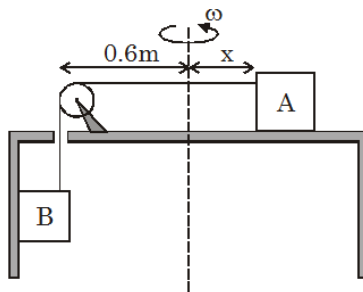
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

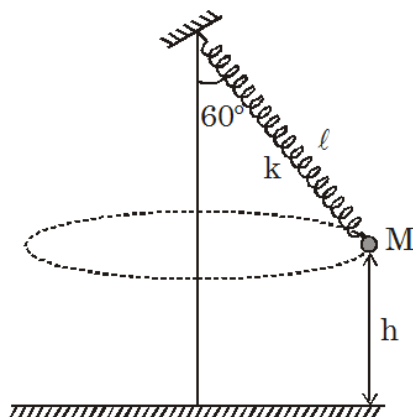
1. Table shown in figure is rotating about its vertical axis with an angular speed $\omega = 2$ rad/sec. Blocks A and B have masses 45 kg and 22.5 kg respectively. String connecting them is light and passes through a small hole in the table. Pulley shown is massless. Coefficient of friction at the contacts of A and B with the table is 0.4. dimensions of two blocks are small and distance of B from rotation axis is 0.6 m. If block A remains stationary relative to the table at a distance 'x' from the rotation axis then (Take : $g = 10$ m/s²)



- (A) Minimum possible value of x is 0.13 m (B) Minimum possible value of x is 0.26 m
(C) Maximum possible value of x is 2.37 m (D) Maximum possible value of x is 1.37 m

Space for Rough Work

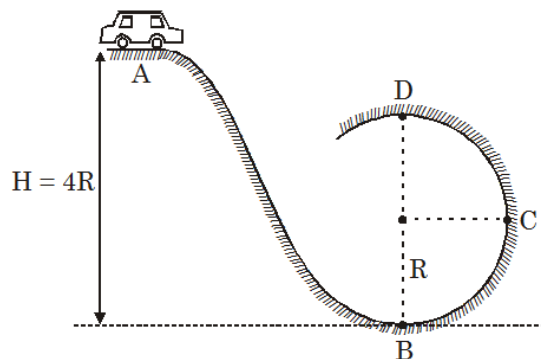
2. A ball of mass M is swing around in a circle in a horizontal plane. One end of spring (spring constant k) is fixed and other end is connected to the ball as shown in figure. The ball describes a horizontal circle at a distance h above the horizontal floor with constant speed. The stretched spring has a length ℓ and makes an angle 60° with the vertical. Neglect air resistance.



- (A) Magnitude of force exerted by spring on block is $2Mg$
- (B) Natural length of spring is $\left(\ell + \frac{2Mg}{k}\right)$.
- (C) Speed of the ball is $\sqrt{\frac{3g\ell}{2}}$
- (D) At some instant of time, if spring breaks, the ball moves a horizontal distance $\sqrt{3h\ell}$, before it hits the floor.

Space for Rough Work

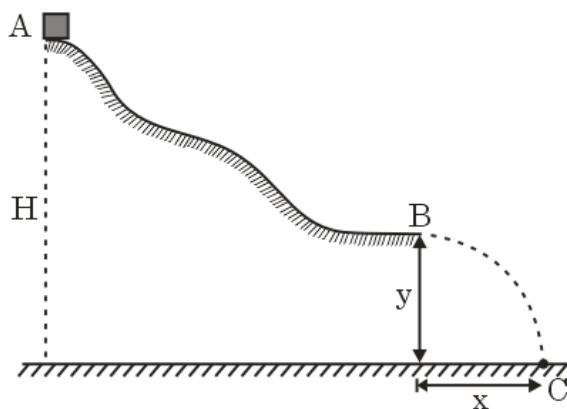
3. A small toy car of mass m slides with negligible friction on a loop track as shown in figure. The toy car starts from rest at a point $H = 4R$ above the lowest point of the track. (Acceleration due to gravity is g and radius of circular part is R).



- (A) Speed of toy car at point B is $\sqrt{4gR}$
- (B) Normal force exerted by track on the toy car at point B is $9\ mg$.
- (C) Normal force exerted by track on the toy car at position C is $6\ mg$.
- (D) Toy car loses contact with track at a point between C and D.

Space for Rough Work

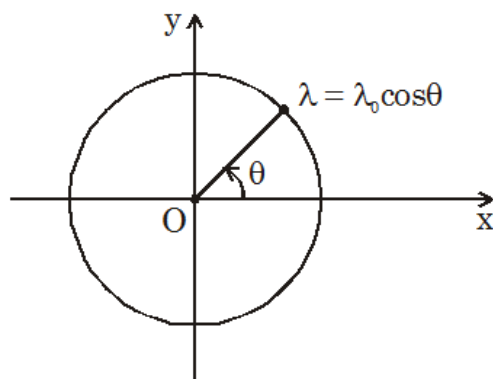
4. A small block slides down a hill (starting from rest) of height H . The part of the hill near its end B is horizontal. The block flies off the hill at B to hit the ground at C . Hill is smooth. If y is height of end B of hill from ground and x is horizontal displacement of block before hitting the ground after losing contact with track, then :



- (A) For x to be maximum, y is H
- (B) For x to be maximum y is $\frac{H}{2}$.
- (C) Maximum possible value of x is $\frac{H}{2}$.
- (D) Maximum possible value of x is H .

Space for Rough Work

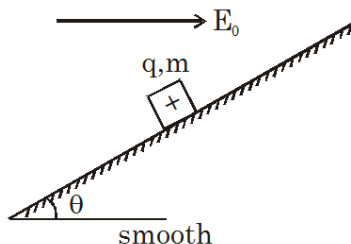
5. A thin non conducting ring of radius R has linear charge density λ varying as $\lambda = \lambda_0 \cos\theta$, where λ_0 is a positive constant and θ is angle measured from positive x-direction as shown. Consider centre of ring is at origin.



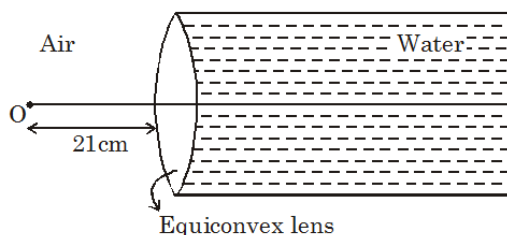
- (A) Resultant electric field at origin is towards positive y-axis.
 (B) Resultant electric field at origin is towards negative x-axis.
 (C) Magnitude of resultant electric field at origin is $\frac{\lambda_0}{8\epsilon_0 R}$.
 (D) Magnitude of resultant electric field at origin is $\frac{\lambda_0}{4\epsilon_0 R}$.

Space for Rough Work

6. A fixed smooth inclined plane making an angle θ with the horizontal is placed in a uniform horizontal electric field E_0 as shown in figure. A small block of mass m having charge $+q$ is released on inclined plane. (Acceleration due to gravity is 'g')



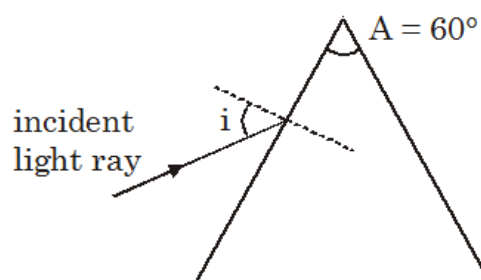
- (A) Normal contact force applied by inclined plane on block is $(mg \cos\theta + qE_0 \sin\theta)$
- (B) Normal contact force force applied by inclined plane on block is $(mg \cos\theta - qE_0 \sin\theta)$
- (C) If acceleration of block is zero then $\theta = \tan^{-1} \left(\frac{qE_0}{mg} \right)$.
- (D) If acceleration of block is zero then $\theta = \tan^{-1} \left(\frac{mg}{qE_0} \right)$.
7. A long cylindrical tube containing water is closed by an equiconvex lens of focal length 10 cm in air. A point object 'O' is placed along the axis of the tube outside it at a distance of 21 cm from the lens. Refractive index of material of lens is 1.5 and that of water is $\frac{4}{3}$.



- (A) Final image is virtual in nature.
- (B) Final image is real in nature.
- (C) Final image is located at a distance 70 cm from lens inside the tube.
- (D) Final image is located at a distance 30 cm from lens inside the tube.

Space for Rough Work

8. A prism with prism angle 60° and refractive index $\sqrt{\frac{7}{3}}$ is given.



- (A) Minimum possible angle of incidence so that the light ray is refracted from the second surface is 30° .
- (B) Minimum possible angle of incidence, so that the light ray is refracted from the second surface is 60° .
- (C) Maximum possible deviation produced by prism is 30° .
- (D) Maximum possible deviation produced by prism is 60° .

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **TWO (02)** List-Match sets.
- Each List-Match set has **TWO** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has Four entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U)
- **Four** options are given in each Multiple Choice Question based on List-I and List-II and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the option corresponding to the correct combination is chosen.
 Zero Marks : 0 If none of the options is chosen (i.e., the question is unanswered)
 Negative Marks : -1 In all other cases

9. A particle is performing SHM $x = A \sin(\omega t + \phi)$ whose time period is T . List-I contains events related to the motion of particle and List-II contains the minimum time after which that event happen. Choose the correct match from option if $\phi = 0^\circ$

	List-I		List-II
(I)	Particle comes to rest	(P)	$T/4$
(II)	Velocity is maximum	(Q)	$T/6$
(III)	Kinetic energy = Potential energy	(R)	$3T/8$
(IV)	Velocity is half of maximum velocity	(S)	$T/24$
		(T)	$T/2$
		(U)	$T/8$

- (A) $I \rightarrow P; II \rightarrow T; III \rightarrow U; IV \rightarrow Q$ (B) $I \rightarrow Q; II \rightarrow P; III \rightarrow R; IV \rightarrow S$
 (C) $I \rightarrow R; II \rightarrow U; III \rightarrow U; IV \rightarrow Q$ (D) $I \rightarrow S; II \rightarrow Q; III \rightarrow R; IV \rightarrow S$

Space for Rough Work

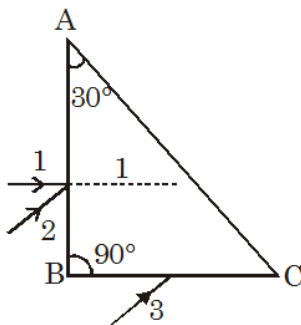
10. A particle is performing SHM $x = A \sin(\omega t + \phi)$ whose time period is T . List-I contains events related to the motion of particle and List-II contains the minimum time after which that event happens. Choose the correct match from option if $\phi = \pi/4$

	List-I		List-II
(I)	Particle comes to rest	(P)	$T/4$
(II)	Velocity is maximum	(Q)	$T/6$
(III)	Kinetic energy = Potential energy	(R)	$3T/8$
(IV)	Velocity is half of maximum velocity	(S)	$T/24$
		(T)	$T/2$
		(U)	$T/8$

- (A) $I \rightarrow P; II \rightarrow T; III \rightarrow U; IV \rightarrow Q$
 (B) $I \rightarrow U; II \rightarrow R; III \rightarrow P; IV \rightarrow S$
 (C) $I \rightarrow R; II \rightarrow Q; III \rightarrow S; IV \rightarrow T$
 (D) $I \rightarrow S; II \rightarrow Q; III \rightarrow R; IV \rightarrow T$

Space for Rough Work

11. ABC is a right-angled prism kept in air. A ray (1) is incident on the face AB along the normal. Refractive index of the material of prism is the minimum value that will be required so that ray (1) undergoes total internal reflection at the face AC. Another ray (2) is incident on the face AB such that it emerges from face AC along the normal to AC. A third ray (3) falls on the face BC and emerges from face AC such that its angle of emergence is the same as that of incidence. Assuming light (1), (2) and (3) have the same wavelength, then match the following.



List-I		List-II	
(I)	Refractive index of the material of prism is	(P)	120
(II)	Angle of incidence in degree of ray (2) is	(Q)	90
(III)	Deviation in degree suffered by ray (2) is	(R)	2
(IV)	Deviation in degree suffered by ray (3) is	(S)	60
		(T)	1.5
		(U)	4

(A) I $\rightarrow$ U; II $\rightarrow$ Q; III $\rightarrow$ S; IV $\rightarrow$ P

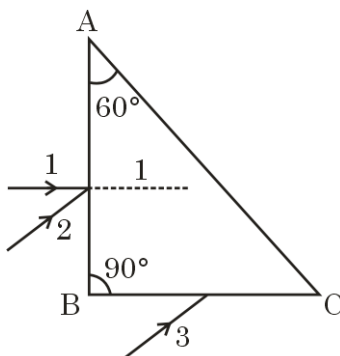
(B) I $\rightarrow$ R; II $\rightarrow$ P; III $\rightarrow$ S; IV $\rightarrow$ P

(C) I $\rightarrow$ U; II $\rightarrow$ S; III $\rightarrow$ P; IV $\rightarrow$ T

(D) I $\rightarrow$ R; II $\rightarrow$ Q; III $\rightarrow$ S; IV $\rightarrow$ P

Space for Rough Work

12. ABC is a right-angled prism kept in air. A ray (1) is incident on the face AB along the normal. Refractive index of the material of prism is the minimum value that will be required so that ray (1) undergoes total internal reflection at the face AC. Another ray (2) is incident on the face AB such that it emerges from face AC along the normal to AC. A third ray (3) falls on the face BC and emerges from face AC such that its angle of emergence is the same as that of incidence. Assuming light (1), (2) and (3) have the same wavelength, then match the following. [Take : $\sin^{-1}(0.29) \approx 17^\circ$, $\sin 15^\circ = 0.258$]



List-I		List-II	
(I)	Refractive index of the material of prism is	(P)	120
(II)	Angle of incidence in degree of ray (2) is	(Q)	90
(III)	Deviation in degree suffered by ray (2) is	(R)	2
(IV)	Deviation in degree suffered by ray (3) is	(S)	30
		(T)	1.15
		(U)	4

- (A) I $\rightarrow$ U; II $\rightarrow$ Q; III $\rightarrow$ S; IV $\rightarrow$ P
 (B) I $\rightarrow$ T; II $\rightarrow$ Q; III $\rightarrow$ S; IV $\rightarrow$ U
 (C) I $\rightarrow$ U; II $\rightarrow$ S; III $\rightarrow$ P; IV $\rightarrow$ T
 (D) I $\rightarrow$ R; II $\rightarrow$ Q; III $\rightarrow$ S; IV $\rightarrow$ P

Space for Rough Work

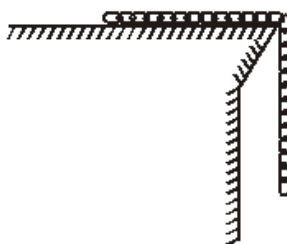
SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

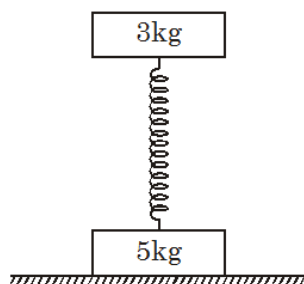
Zero Marks : 0 In all other cases.

1. A uniform chain of mass M and length L rests on a rough horizontal table such that one of its ends hangs over the edge. The chain begins to slide if length of overhanging part is at least equal to two third of the total length of the chain. Chain is released and starts its motion from a position where two third of its total length is overhanging. If work done by friction force acting on the chain by the moment it completely slides off the table is given by $\frac{-\mu MgL}{N}$ then find the value of N . Coefficient of friction between chain and rough horizontal surface is μ and acceleration due to gravity is g .

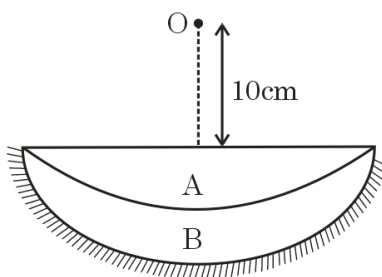


Space for Rough Work

2. The ends of a spring are attached to two blocks of masses 5 kg and 3 kg as shown in figure. The 5 kg block rests on a horizontal surface and the 3 kg block which is vertically above it is in equilibrium with the spring compressed by 1 cm. The 3 kg block is pushed down further by x cm and released. Find the minimum value of x so that the 5 kg block may be lifted off the ground during the subsequent motion.

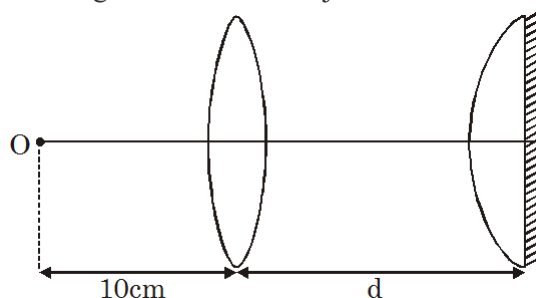


3. A thin plano convex lens A has material of refractive index $\mu_A = 1.8$ and its curved surface has radius of curvature R . A thin layer of transparent material B is laid over the curved surface of A. The refractive index of B is 1.2 and the curved surface of B that is not touching A has radius of curvature $\frac{R}{2}$. This surface of B is silvered. A point object 'O' is kept at a distance of 10 cm on the principal axis of the system (above the plane surface) and its image is formed at a distance of 40 cm above the plane surface. Find R (in cm).



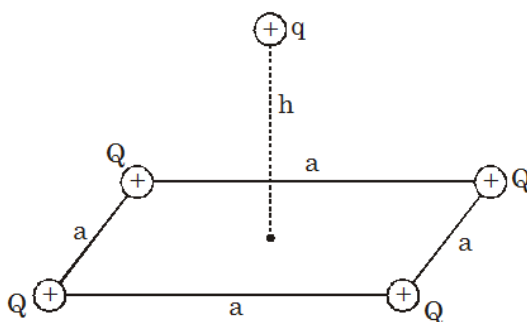
Space for Rough Work

4. A convex lens of focal length 20 cm and another plano-convex lens of focal length 40 cm are placed co-axially. The plane surface of the plano convex lens is silvered. An object O is kept on the principal axis at a distance of 10 cm from the convex lens as shown in figure. Find the distance d (in cm) between two lenses so that final image is formed on object itself.



5. Four small particles charged with equal positive charge $+Q$ each are arranged at the four corners of a horizontal square of side a . A particle having charge $+q$ and mass m is placed at a height h above the centre of the square. Condition under which particle at height h is in equilibrium is given by

$$m = \frac{Qqh}{g\pi\epsilon_0 \left(h^2 + \frac{a^2}{N_1} \right)^{N_2}}. \text{ Fill the value of } (N_1 + N_2). \text{ (Acceleration due to gravity is 'g')}$$



6. A particle performs SHM along x-axis according to the relation $x = 3 \sin(100t) + 8 \cos^2(50t)$ where 'x' is in cm and 't' is in second. Find the maximum possible distance (in cm) of the particle from origin during its motion.

Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

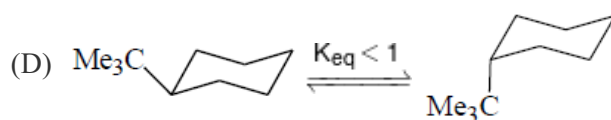
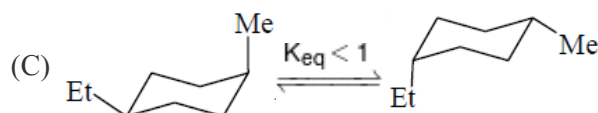
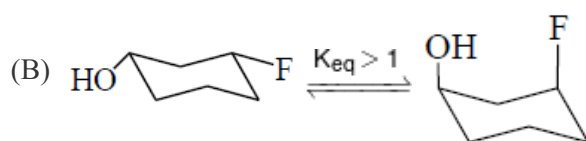
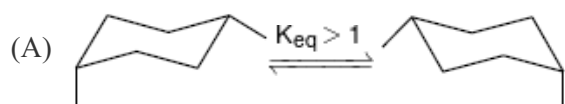
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

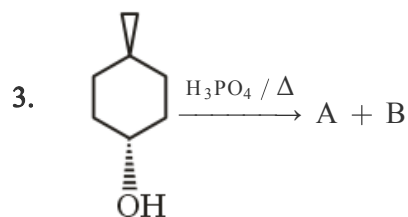
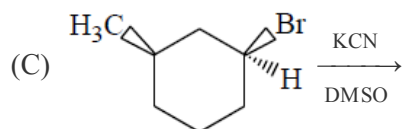
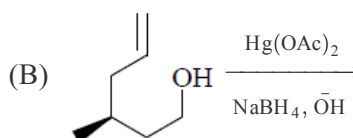
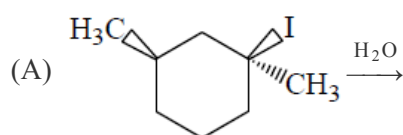
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

1. Which of the following is/are correctly match with their equilibrium constant (K_{eq})



Space for Rough Work

2. Which of the following reaction result the formation of a pair of diastereomers?



Identify relation between A and B. If both A and B give Bromine water test

- (A) Stereoisomers (B) Structural isomers
(C) Enantiomers (D) Diastereomers
4. Choose the correct statement among the following :
- (A) Primitive cubic unit cell involves two dimensional hexagonal closed packed array of atoms
(B) Four hexagonal closed packed two dimensional layers of atoms are involved in one FCC unit cell
(C) Three hexagonal close packed two dimensional layers of atoms are involved in one HCP unit cell
(D) Four square close packed two dimensional layers of atoms are involved in one BCC unit cell

Space for Rough Work

5. Hydrocarbon contains $\frac{600}{7}\%$ by mass carbon and its one molecule weighs 93.33×10^{-24} g.

Which of the following statement(s) is/are correct $[N_A = 6.0 \times 10^{23}]$

- (A) Empirical formula is CH (B) Empirical formula is CH_2
(C) Molecular formula of C_3H_8 (D) Molecular formula is C_4H_8
6. Which of the following given order is **CORRECT** with respect to bond angle?

- (A) $\text{Cl}-\overset{\text{A}}{\text{Se}}-\text{Cl} > \text{Cl}-\overset{\text{A}}{\text{Se}}-\text{O}$ (In SeOCl_2)
(B) $\text{F}_{\text{eq}}-\overset{\text{A}}{\text{S}}-\text{F}_{\text{eq}}$ (in SF_4) $> \text{F}_{\text{eq}}-\overset{\text{A}}{\text{Sb}}-\text{F}_{\text{eq}}$ (In SbF_4^-)
(C) $\text{F}_{\text{eq}}-\overset{\text{A}}{\text{S}}-\text{F}_{\text{eq}}$ (in SF_4) $< \text{F}_{\text{eq}}-\overset{\text{A}}{\text{Sb}}-\text{F}_{\text{eq}}$ (In SbF_4^-)
(D) $\text{Cl}-\overset{\text{A}}{\text{Se}}-\text{Cl} < \text{Cl}-\overset{\text{A}}{\text{Se}}-\text{O}$ (In SeOCl_2)

7. Which of the following species is/are planar and polar?

- (A) NO_2 (B) ClO_2 (C) XeF_5^- (D) ClF_3

8. Choose the **INCORRECT** order of the indicated property given in the options.

(A)	$\text{He} < \text{Ne} < \text{Kr} < \text{Ar}$	:	Clathrate compound formation tendency order
(B)	$\text{NaO}_2 < \text{KO}_2 < \text{RbO}_2 < \text{CsO}_2$	:	Thermal stability order
(C)	$\text{BeF}_2 < \text{BaF}_2$	:	Solubility order
(D)	$\text{LiF} > \text{LiCl} > \text{LiI}$	:	Melting point order

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **TWO (02)** List-Match sets.
- Each List-Match set has **TWO** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has Four entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U)
- **Four** options are given in each Multiple Choice Question based on List-I and List-II and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the option corresponding to the correct combination is chosen.
 Zero Marks : 0 If none of the options is chosen (i.e., the question is unanswered)
 Negative Marks : -1 In all other cases

9. Answer the following by approximately matching the lists based on the information given in the paragraph.

For an n^{th} order reaction $A \rightarrow \text{products}$,

The rate law is $\frac{-dC_A}{dt} = kC_A^n$. Where C_A = concentration of A at time 't'. Let C_{A_0} is initial concentration of A and α = degree of dissociation of reactant 'A'

	List-I		List-II
(I)	$n = 0$	(P)	$\alpha = \frac{C_{A_0} - C_A}{C_{A_0}}$
(II)	$n = \frac{1}{2}$	(Q)	$\alpha = 1 - e^{-kt}$
(III)	$n = 1$	(R)	$\alpha = \frac{kt}{C_{A_0}}$
(IV)	$n = 2$	(S)	$\alpha = \frac{C_{A_0}kt}{(1 + C_{A_0}kt)}$
		(T)	α = independent of initial concentration of reactant
		(U)	Graph between ' α ' and 't' is a straight line passing through origin

Which of the following options has the correct combination considering List-I and List-II ?

- (A) II $\rightarrow$ P, R (B) I $\rightarrow$ P,R,T (C) III $\rightarrow$ P,Q,U (D) III $\rightarrow$ P,Q,T

Space for Rough Work

10. Answer the following by approximately matching the lists based on the information given in the paragraph.

For an n^{th} order reaction $A \rightarrow \text{products}$,

The rate law is $\frac{-dC_A}{dt} = kC_A^n$. Where C_A = concentration of A at time 't'. Let C_{A_0} is initial concentration of A and α = degree of dissociation of reactant 'A'

	List-I		List-II
(I)	$n = 0$	(P)	$\alpha = \frac{C_{A_0} - C_A}{C_{A_0}}$
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(III)	$n = 1$	(R)	$\alpha = \frac{kt}{C_{A_0}}$
(IV)	$n = 2$	(S)	$\alpha = \frac{C_{A_0} kt}{(1 + C_{A_0} kt)}$
		(T)	α = independent of initial concentration of reactant
		(U)	Graph between ' α ' and 't' is a straight line passing through origin

Which of the following options has the correct combination considering List-I and List-II ?

- (A) I $\rightarrow$ P,Q (B) IV $\rightarrow$ S,T (C) IV $\rightarrow$ P,S (D) III $\rightarrow$ Q,U

Space for Rough Work

11.

List-I Dimerization/polymerization by			List-II
(I)	$3c - 2e^-$	(P)	BeX_2
(II)	Co-ordinate bond	(Q)	AlX_3
(III)	Odd e^-	(R)	CX_3
(IV)	No dimerization	(S)	BX_3
		(T)	IX_3
		(U)	ClO_3

 Select correct match when $X = F$.

- (A) $II \rightarrow R$ (B) $I \rightarrow Q$ (C) $III \rightarrow U$ (D) $IV \rightarrow S$

12.

List-I Dimerization/polymerization by			List-II
(I)	$3c - 2e^-$	(P)	BeX_2
(II)	Co-ordinate bond	(Q)	AlX_3
(III)	Odd e^-	(R)	CX_3
(IV)	No dimerization	(S)	BX_3
		(T)	IX_3
		(U)	ClO_3

 Select correct match when $X = Cl$

- (A) $I \rightarrow P$ (B) $III \rightarrow R$ (C) $II \rightarrow Q$ (D) $IV \rightarrow T$

Space for Rough Work

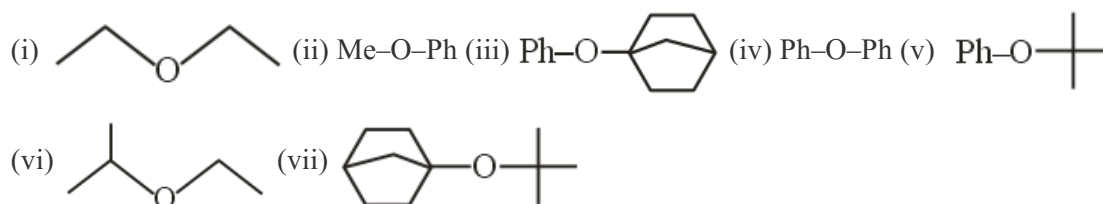
SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

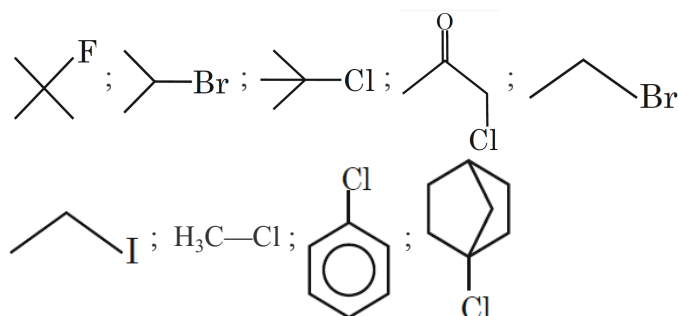
Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. Number of following compounds which can't be synthesised in good yield by Williamson's reaction

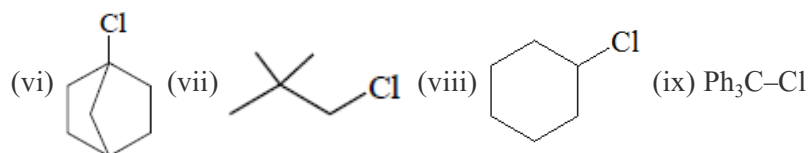
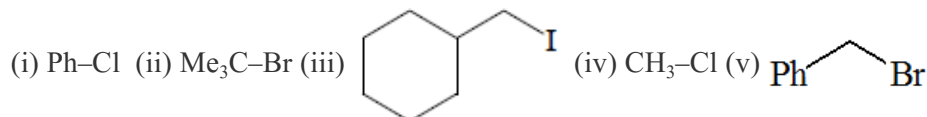


2. Number of compounds which are more reactive than $\text{CH}_3-\underset{\text{CH}_3}{\text{CH}}-\text{Cl}$ by S_N^2 mechanism?



Space for Rough Work

3. How many of the following substrate can give E2 elimination reaction on treatment with base / Δ



4. What should be the minimum pressure (in atm) applied on 0.1 M glucose solution, filled in semi-permeable membrane, so as to filter away 750 mL water, leaving 250 mL of glucose solution at 300 K.
[Given : $R = 0.08 \text{ atm LK}^{-1} \text{ mol}^{-1}$]
5. An aq. solution of 0.5M KMnO_4 is divided into two parts. One part of it requires 125 ml of 1.5M aq. solution of oxalate ions in acidic medium, while another part requires 270 ml of 0.5M aq. solution of iodide ions in neutral medium which are converted into I_2 only. Calculate total volume (mL) of the initial KMnO_4 solution.
6. If s and p-orbital mixing (s-p mixing) is operative for all species given, then how many following species is/are paramagnetic in nature ?
 O_2^- , N_2 , C_2 , CN^- , B_2 , O_2^{2+}

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

1. Consider $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfying the equation $f(x + y) = f(x - y) + f(x).f(y) \forall x, y \in \mathbb{R}$.

Identify the correct statement(s)

(A) $f(2014) + f(2015) = f(2016)$

(B) $\sin^{-1}(f(2)) + \cos^{-1}(f(3)) = \frac{\pi}{2}$

(C) $f(2014) - f(2015) = f(2016)$

(D) $\sin^{-1}(f(3)) + \cos^{-1}(f(2)) = \frac{\pi}{2}$

Space for Rough Work

2. ABC is a triangle in which $\angle C = 2\angle B$. D is a point on BC such that AD bisects $\angle BAC$ and $AC = BD$. Then which of the following is/are true

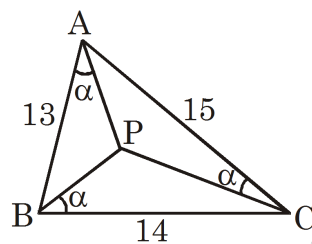
(A) $\cos B + \cos C = \frac{\sqrt{5}}{2}$

(B) $\cos B - \cos C = \frac{1}{2}$

(C) $\frac{\sin A}{\sin C} = 1$

(D) $\frac{\sin A}{\sin C} = \frac{\sqrt{5} + 1}{4}$

3. For the figure given below if $\tan \alpha = \frac{m}{n}$, where m and n are relatively prime positive integers then



(A) $m + n = 463$

(B) $n - m = 122$

(C) $2m - n = 41$

(D) $m + n = 465$

Space for Rough Work

4. The value of $\lim_{t \rightarrow 1^-} (1-t) \sum_{r=1}^{\infty} \frac{t^r}{1+t^r}$ is

(A) $\ln 2$

(B) $\ln 3$

(C) $\ln \left(\frac{2e}{1+e} \right)$

(D) $\frac{1}{2}$

5. Which of the following are even functions in their domain

(A) $y(x) = \frac{(2 \sin x - 1)(\sin^2 x - 1)}{(2 \sin x - 1)}$

(B) $y(x) = f \circ g \circ h(x)$ (where f is neither even nor odd, g is even and h is an odd function, all functions are defined for all $x \in \mathbb{R}$)

(C) $y(x)$ is defined for all real x and $y(x) = |x - 1|$ for $x \in [0, 2]$, and $y(x)$ is periodic with period 2.

(D) $y(x) = \frac{1 + e^{x-1} (\cos(1-x) + \cos(1+x)) + e^{2x}}{e^x}$

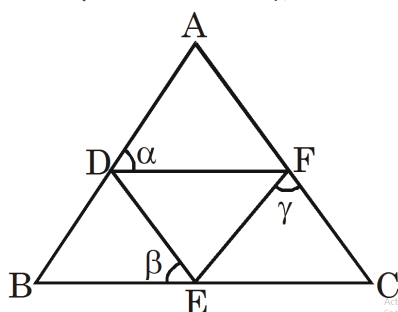
Space for Rough Work

6. For the equation $\cos^{-1} \left(\frac{x^2 - 1}{x^2 + 1} \right) + \sin^{-1} \left(\frac{2x}{x^2 + 1} \right) + \tan^{-1} \left(\frac{2x}{x^2 - 1} \right) = \frac{2\pi}{3}$,

which of the following is/are correct.

- (A) The number of solution is 2 (B) It has at least one irrational solution
(C) The sum of solution is positive (D) The sum of solution is negative

7. In the diagram shown, AB and AC are equal sides of an isosceles triangle ABC. The equilateral triangle DEF is inscribed. If $\angle ADF$ be α , $\angle BED$ be β and $\angle CFE$ be γ , then **CORRECT** relation is/are



- (A) $\beta = \frac{\alpha + \gamma}{2}$ (B) $\alpha = \frac{\beta + \gamma}{2}$ (C) $\beta = \frac{\alpha - \gamma}{2}$ (D) $\alpha = \frac{\beta - \gamma}{2}$

8. If α ($\alpha > 0$) is a root of $x^{11} - x^{10} + x^8 - x^7 + x^5 - x^4 + x^2 - x - 20 = 0$ then α^{12} is greater than

- (A) 59 (B) 61 (C) 58 (D) 57

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **TWO (02)** List-Match sets.
- Each List-Match set has **TWO** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has Four entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U)
- **Four** options are given in each Multiple Choice Question based on List-I and List-II and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the option corresponding to the correct combination is chosen.
 Zero Marks : 0 If none of the options is chosen (i.e., the question is unanswered)
 Negative Marks : -1 In all other cases

9. Match List-I with List-II and select the correct answer using the code given below the list.

List-I Equations		List-II No. of solutions	
(I)	$x^2 \tan x = 1 \quad x \in \left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$	(P)	5
(II)	$2^{\cos x} = \sin x , x \in [0, 2\pi]$	(Q)	1
(III)	If $f(x)$ is a polynomial of degree 5 with real coefficients such that $f(x) = 0$ has 8 real roots then, roots of $f(x) = 0$	(R)	3
(IV)	$7^{ x } (5 - x) = 1$	(S)	4
		(T)	0
		(U)	10

Which of the following is the only **CORRECT** combination ?

- (A) I $\rightarrow$ P (B) II $\rightarrow$ Q (C) III $\rightarrow$ R (D) IV $\rightarrow$ S

Space for Rough Work

10. Match List-I with List-II and select the correct answer using the code given below the list.

List-I Equations		List-II No. of solutions	
(I)	$x^2 \tan x = 1 \quad x \in \left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$	(P)	5
(II)	$2^{\cos x} = \sin x , x \in [0, 2\pi]$	(Q)	1
(III)	If $f(x)$ is a polynomial of degree 5 with real coefficients such that $f(x) = 0$ has 8 real roots then, roots of $f(x) = 0$	(R)	3
(IV)	$7^{ x } (5 - x) = 1$	(S)	4
		(T)	0
		(U)	10

Which of the following is the only **INCORRECT** combination ?

- (A) $I \rightarrow R$ (B) $II \rightarrow S$ (C) $III \rightarrow P$ (D) $IV \rightarrow Q$

Space for Rough Work

11. Match List-I with List-II and select the correct answer using the code given below the list.

List-I		List-II	
(I)	Number of solutions of $4(\sin^6\theta + \cos^6\theta) = 10 - 18\sin\theta\cos\theta$ in $[0, 3\pi]$, is	(P)	1
(II)	Number of possible pairs (x, y) satisfying $x^2 - 6x + 10 - \sin^2y = 0$, where $y \in [0, 2\pi]$ and $x \in \mathbb{R}$, is	(Q)	2
(III)	Number of integral values of x satisfying $\sin x < \frac{1}{2}$ in $[0, 2\pi]$, is	(R)	3
(IV)	Number of solutions satisfying $\tan x + \sec x = 2\cos x$ in $[0, 4\pi]$, is	(S)	4
		(T)	5
		(U)	6

Which of the following is the only **CORRECT** combination ?

- (A) $I \rightarrow Q$ (B) $I \rightarrow R$ (C) $II \rightarrow P$ (D) $II \rightarrow R$

Space for Rough Work

12. Match List-I with List-II and select the correct answer using the code given below the list.

List-I		List-II	
(I)	Number of solutions of $4(\sin^6\theta + \cos^6\theta) = 10 - 18\sin\theta\cos\theta$ in $[0, 3\pi]$, is	(P)	1
(II)	Number of possible pairs (x, y) satisfying $x^2 - 6x + 10 - \sin^2y = 0$, where $y \in [0, 2\pi]$ and $x \in \mathbb{R}$, is	(Q)	2
(III)	Number of integral values of x satisfying $\sin x < \frac{1}{2}$ in $[0, 2\pi]$, is	(R)	3
(IV)	Number of solutions satisfying $\tan x + \sec x = 2\cos x$ in $[0, 4\pi]$, is	(S)	4
		(T)	5
		(U)	6

Which of the following is the only **CORRECT** combination ?

- (A) III $\rightarrow$ S (B) III $\rightarrow$ U (C) IV $\rightarrow$ T (D) IV $\rightarrow$ S

Space for Rough Work

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. Let $S_n = \sum_{k=0}^n \tan^{-1} \frac{49}{49k^2 + 63k + 57}$ then $\cot\left(\lim_{n \rightarrow \infty} S_n\right)$ is equal to
2. In a triangle ABC, medians AD and CE are drawn. If $AD=5$, $\angle DAC = \frac{\pi}{8}$ and $\angle ACE = \frac{\pi}{4}$ then the area of the triangle ABC is equal to
3. If $\lim_{x \rightarrow 0^+} \left(\frac{\sin^2 x}{\left[\frac{x}{\pi} \right] + \frac{x^2}{\pi^2}} \right) \times \frac{\sin(\sin x) - \sin x}{ax^5 + bx^3 + c} = \frac{-\pi^2}{12}$. Then value of $b + c$ is
(Where $[.]$ denotes greatest integer function)

Space for Rough Work

4. If the value of $\lim_{x \rightarrow \infty} x^2 \tan \left(\ln \left(\sqrt{\cos \frac{\pi}{x}} \right) \right)$ is equal to $-k\pi^2$, then k is equal to
5. Let $f(\theta) = \frac{1}{\tan^9 \theta} \left((1 + \tan \theta)^{10} + (2 + \tan \theta)^{10} + \dots + (20 + \tan \theta)^{10} \right) - 20 \tan \theta$. The value of $\lim_{\theta \rightarrow \frac{\pi}{2}^-} f(\theta)$ is
6. The number of solution of equation $2\sin 2x + \sin x (1 - 2\cos 2x) - 4 = 0$ in the interval $(0, \pi)$:
-

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

Space for Rough Work